FINAL PROJECT · TRIPLETEN DATA ANALYST CERTIFICATION

Delay, promise and product: what explains negative reviews at Olist

Luciana Jost · October 2026

Data: Brazilian E-Commerce Public Dataset by Olist (Kaggle) · 95,560 delivered and reviewed orders · purchases from Jan 2017 to Aug 2018

CONTEXT

One in eight reviews is negative

12.8%

of delivered orders get a score of 1 or 2, the main indicator of the project

Question

Which factors in the purchase journey reduce Olist's customer satisfaction, and which one weighs the most?

Data

95,560 delivered and reviewed orders, purchased between January 2017 and August 2018, with score, comment, dates, state and category

Source: Olist, Brazilian E-Commerce Public Dataset (Kaggle). Own analysis.

CONTEXT

Four hypotheses about where dissatisfaction comes from

H1 · Broken promise

With a similar wait, late orders get worse reviews. Basis: expectation disconfirmation (Oliver, 1980).

H2 · Reviewing without the product

Among late orders, customers who answer the survey before receiving the order review worse.

H3 · Loss aversion

Each day late lowers the score more than each day early raises it (Kahneman and Tversky, 1979).

H4 · Product

In negative reviews of on-time orders, product complaints weigh more than in those of late orders.

APPROACH

How the hypotheses were tested

1

Preparation

7 tables joined into one row per order; cutoff from Jan 2017 to Aug 2018

2

Tests

z-test, Mann-Whitney and Fisher; α = 5% with Bonferroni; comparisons within waiting time bands

3

Comments

9,194 negative comments classified by keywords and checked by hand on 100

4

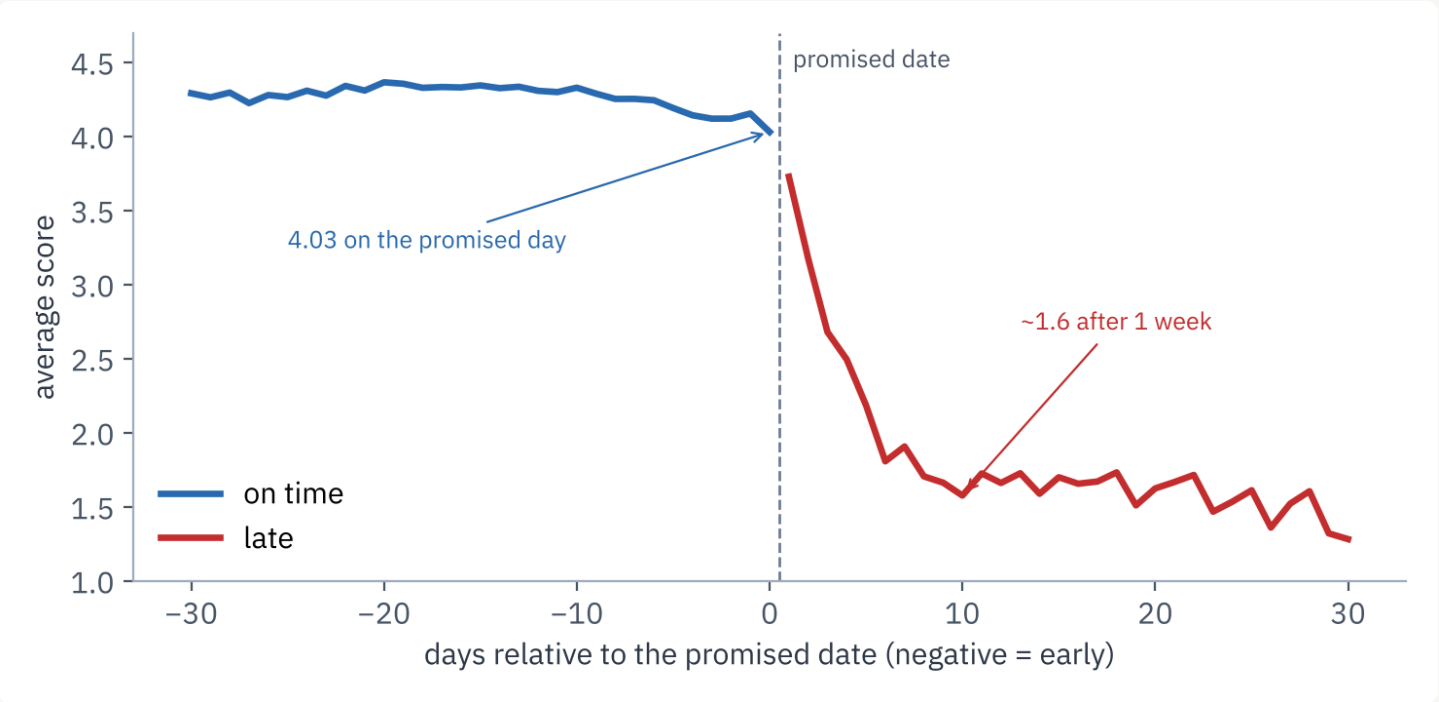
Dashboard

Tableau Public with the KPIs and filters for period, state and category

Observational data: the results indicate association, not causation.

RESULT · H1 SUPPORTED

Being late weighs more than waiting



+24 to 53 pp

negative reviews on late orders, compared with on-time orders with a similar wait

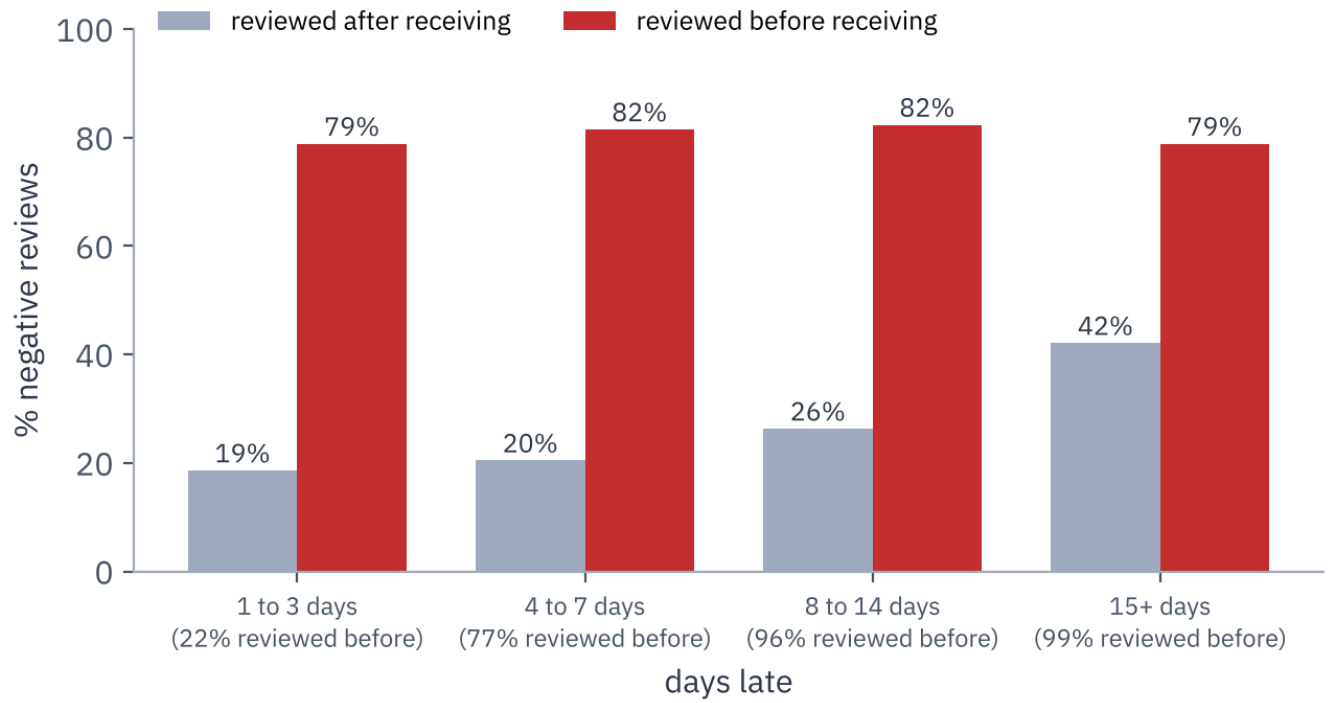
+14 pp

effect of waiting more than 30 days, within the promised date

z-test and Mann-Whitney by delivery time band, adjusted α of 0.01. The result holds within São Paulo.

RESULT · H2 SUPPORTED

70% of late customers review before receiving the order



The survey is sent on the **estimated date**. When the order is late, it arrives before the product.

Customers who review without having received it give a 1 or 2 in about **80%** of cases, whatever the size of the delay.

With the product in hand, the rate stays between 19% and 26% for delays of up to 14 days.

Association: the customer chooses when to answer. z-test and Mann-Whitney by delay band, adjusted α of 0.0125.

RESULT · H3 NOT SUPPORTED AS FORMULATED

The asymmetry disappears once the customer has the product

12×

score lost per day late compared with the gain per day early, in the full table

0.9×

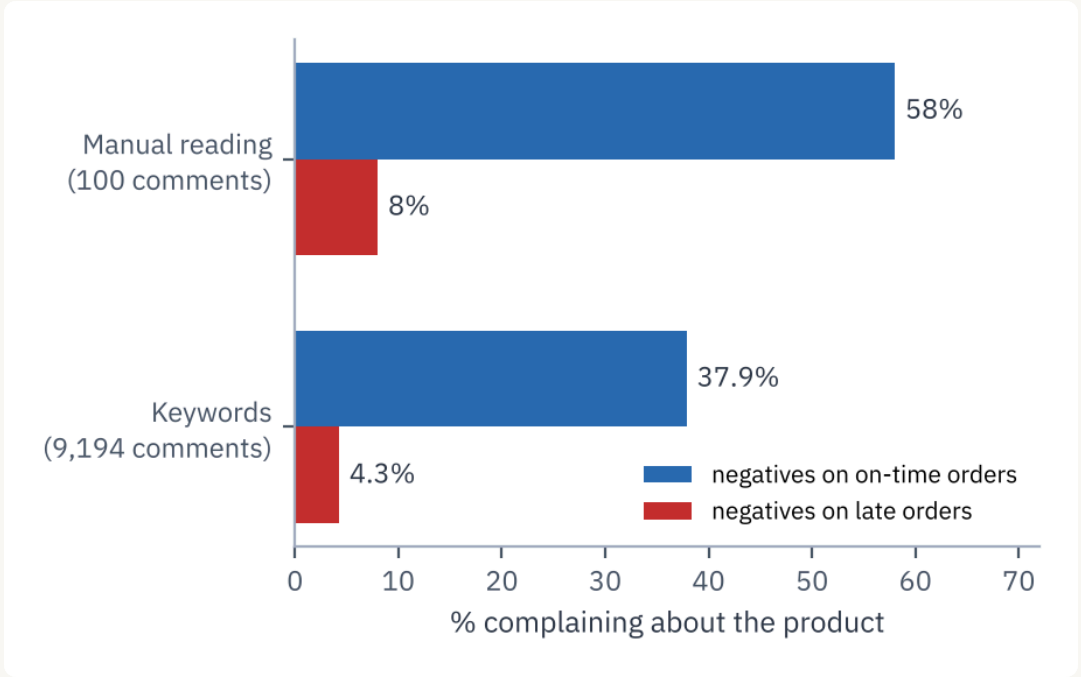
the same ratio among those who reviewed after delivery: difference not significant

The slope came from H2: with each day of delay, more customers review without the product. With the product in hand, the effect lies in crossing the promised date (up to +5.5 pp), not in each extra day.

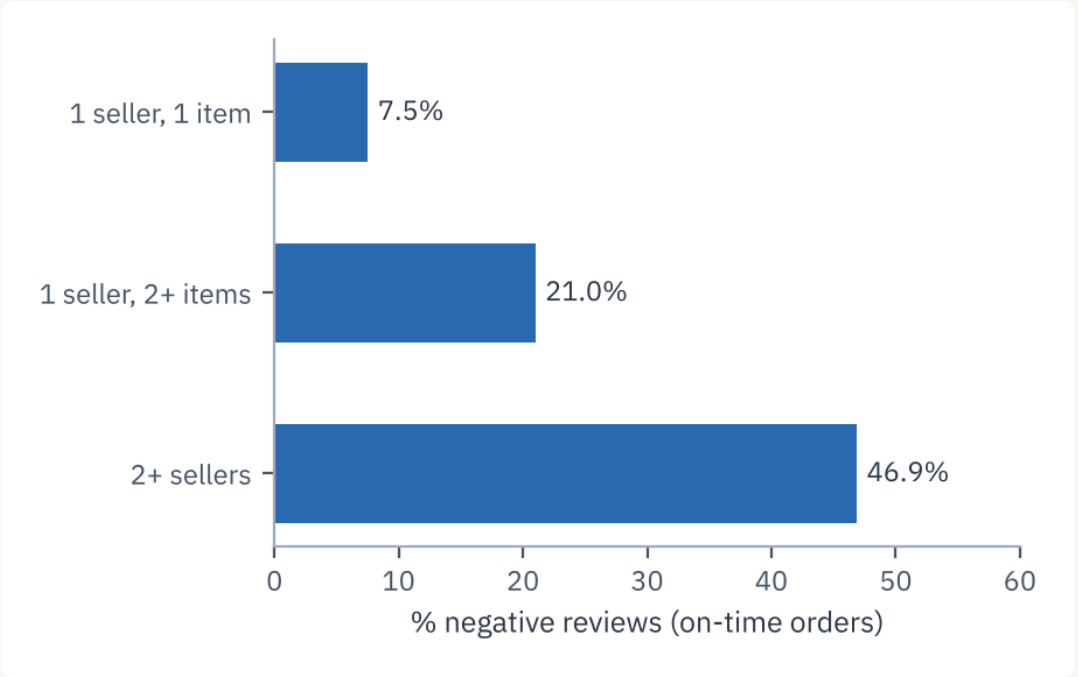
RESULT · H4 SUPPORTED AND COMPLEMENTARY ANALYSIS

On time, dissatisfaction comes from the product and from split orders

Product complaints in negative reviews

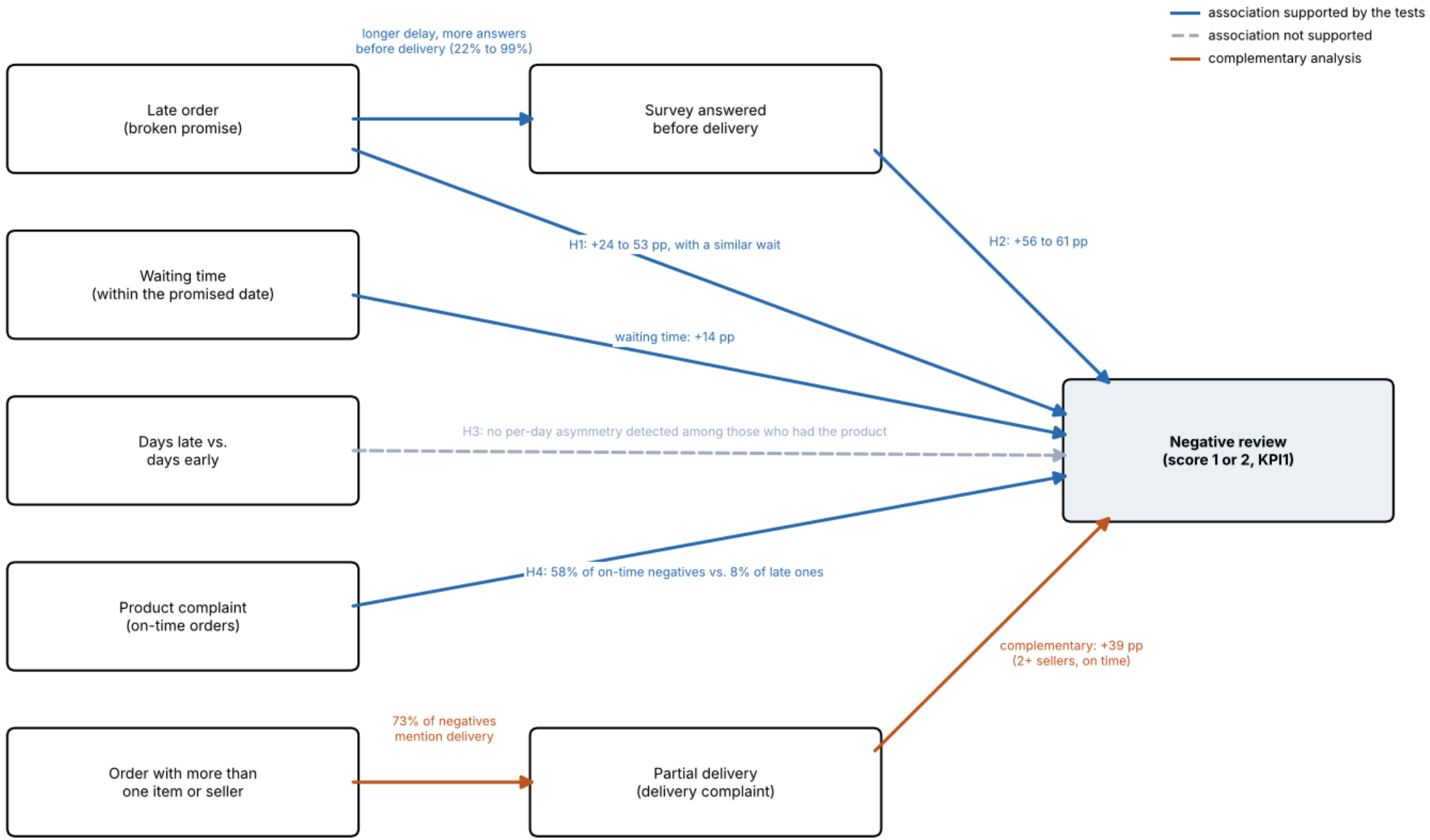


Negative reviews on orders delivered on time



Keywords checked on 100 comments: 94% precision and 48% recall, with similar error in both groups. Fisher $p = 1.1 \times 10^{-7}$.

Association model for negative reviews



RESULT · SYNTHESIS

In practice: three sources of dissatisfaction

1. Delay	Order delivered after the promised date	→ Survey answered before delivery (70% of late orders)	→ **+56 to 61 pp negative reviews (H1, H2)**
2. Product	Simple order delivered on time	→ Product complaint (58% of on-time negatives)	→ **Sellers and listings (H4)**
3. Split order	Order with more than one seller	→ Partial delivery (73% of negatives mention delivery)	→ **+39 pp negative reviews, even on time**

H3 not supported: among those who reviewed with the product in hand, one more day of delay does not lower the score in a detectable way. Arrows indicate association, not cause.

RESULT · RECOMMENDATIONS

Five lines of action, each backed by its evidence

Recommendation	Basis	Source
R1. Send the survey after delivery is confirmed, with an A/B test: about 385 late orders per group, around 2.5 months	H2 and H3	1. Delay
R2. Warn customers about a delay before the promised date, with a new date	H1	1. Delay
R3. Prioritize logistics in AL, MA, SE and RJ	States, H1	1. Delay
R4. Record delivery per item and notify when the order arrives in parts	Composition	3. Split order
R5. Monitor product complaints by seller and category	H4	2. Product

RESULT · CAVEATS

Limitations and next steps

Limitations

- Observational data: association, not cause
- Data from 2017 and 2018; the operation may have changed
- Customers choose when to answer (H2 and H3)
- 48% recall: keyword rates are a lower bound
- A single delivery date per order

Next steps

- Run the R1 A/B test and measure the causal effect
- Expand the term list and validate on a new sample
- Cross product complaints with sellers
- Repeat the analysis with recent operating data

Project materials

Interactive dashboard (Tableau Public)
public.tableau.com/views/OlistExperinciadoclienteeavaliaonegativa/PainelOlist

Notebook, data and presentation (GitHub)
github.com/lucianajost/data_analytics_portfolio